第四章 需求规格文档

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主要内容

1. 需求规格说明概述
2. 需求规格说明文档
3. 模版的选择与裁剪
4. 文档写作技巧
5. 优秀需求规格说明文档的特性
6. 需求规格说明的实践调查

6. 需求规格说明的实践调查

- 需求规格说明文档的内容

<table>
<tr><td rowspan="11">问题域描述</td><td colspan="2">业务过程</td></tr>
<tr><td colspan="2">操作</td></tr>
<tr><td colspan="2">功能</td></tr>
<tr><td colspan="2">用户行为</td></tr>
<tr><td colspan="2">任务</td></tr>
<tr><td colspan="2">事件</td></tr>
<tr><td colspan="2">场景</td></tr>
<tr><td>术语</td><td>首字母缩写</td></tr>
<tr><td>量（Volume）</td><td>估计值</td></tr>
<tr><td colspan="2">公司背景</td></tr>
</table>

6. 需求规格说明的实践调查

- 需求规格说明文档的内容

效果（解系统描述）	特征	通用标准特征
		独特特征
	事务	更新
		插入
		删除
		修改
	信息需求	特定报告（Ad hoc reporting）
		数据采集
		数据流
		数据库
		查询处理
		报表
	行为需求	困难示例（Corner case）
		错误示例（Error case）
		事件
		外部事件
		状态
		转移
	转换/转移（Transformation）需求	

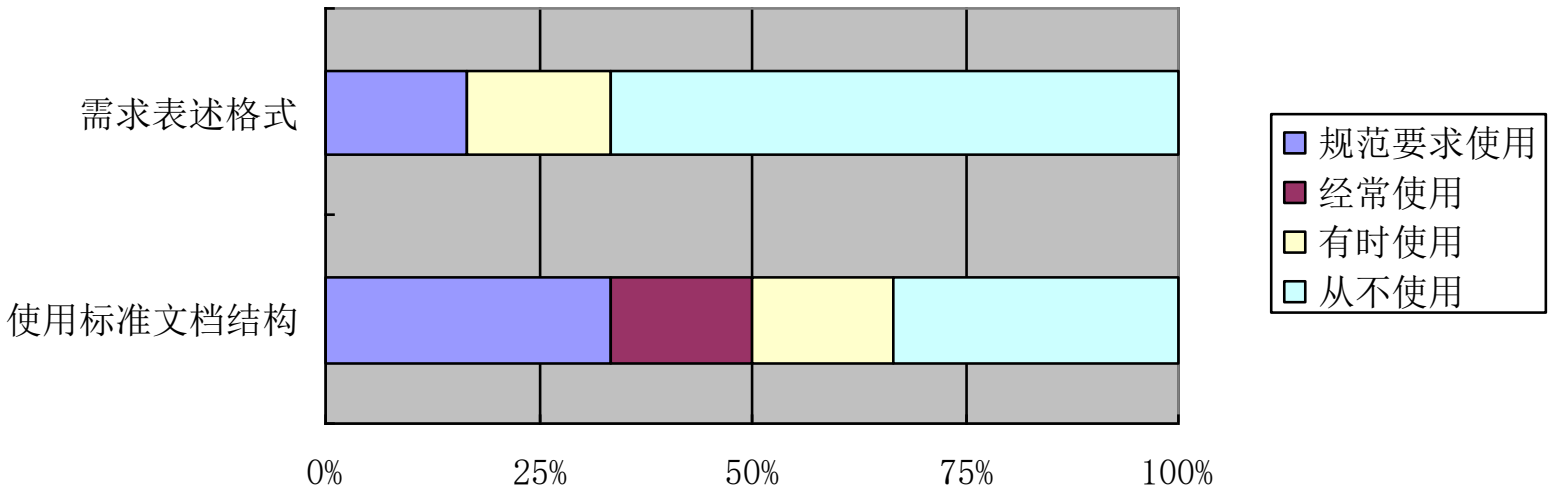
6. 需求规格说明的实践调查

- 需求规格说明文档的内容

<table>
<tr><td colspan="3">问题</td></tr>
<tr><td rowspan="3">接口</td><td colspan="2">与其他系统接口</td></tr>
<tr><td colspan="2">系统接口</td></tr>
<tr><td colspan="2">用户界面</td></tr>
<tr><td colspan="3">变更</td></tr>
<tr><td>目的</td><td colspan="2">目标</td></tr>
<tr><td colspan="3">可行性分析</td></tr>
<tr><td colspan="3">架构</td></tr>
<tr><td colspan="3">约束</td></tr>
<tr><td rowspan="6">文档信息</td><td rowspan="2">文档历史</td><td>版本和草案</td></tr>
<tr><td>签署日期</td></tr>
<tr><td>传 播 （ Circulation ）</td><td>授权列表</td></tr>
<tr><td colspan="2">原创作者</td></tr>
<tr><td colspan="2">目录</td></tr>
<tr><td colspan="2">参考文献</td></tr>
</table>

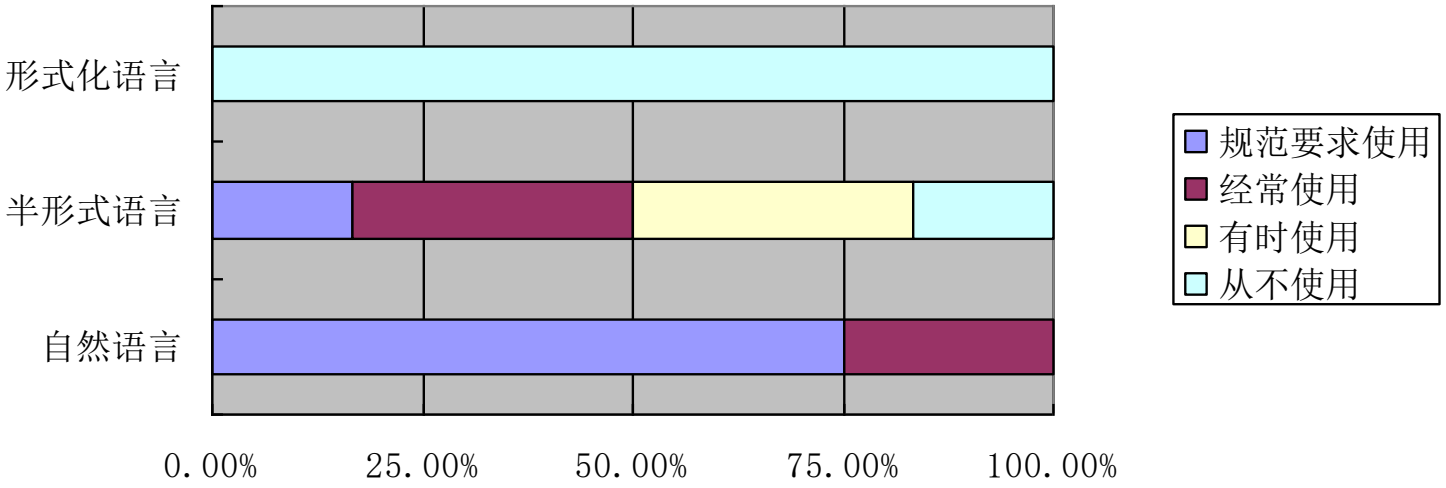
6. 需求规格说明的实践调查

- 示例和模板的使用



6. 需求规格说明的实践调查

- 需求规格说明文档的描述语言



IEEE 830标准（英文版）(1)

IEEE Recommended Practice for Software Requirements Specifications

1. Overview

This recommended practice describes recommended approaches for the specification of software requirements. It is divided into five clauses. Clause 1 explains the scope of this recommended practice. Clause 2 lists the references made to other standards. Clause 3 provides definitions of specific terms used. Clause 4 provides background information for writing a good SRS. Clause 5 discusses each of the essential parts of an SRS. This recommended practice also has two annexes, one which provides alternate format templates, and one which provides guidelines for compliance with IEEE/EIA 12207.1-1997.

1.1 Scope

This is a recommended practice for writing software requirements specifications. It describes the content and qualities of a good software requirements specification (SRS) and presents several sample SRS outlines.

This recommended practice is aimed at specifying requirements of software to be developed but also can be applied to assist in the selection of in-house and commercial software products. However, application to already-developed software could be counterproductive.

When software is embedded in some larger system, such as medical equipment, then issues beyond those identified in this recommended practice may have to be addressed.

This recommended practice describes the process of creating a product and the content of the product. The product is an SRS. This recommended practice can be used to create such an SRS directly or can be used as a model for a more specific standard.

This recommended practice does not identify any specific method, nomenclature, or tool for preparing an SRS.

IEEE 830标准（英文版）(2)

2. References

This recommended practice shall be used in conjunction with the following publications.

ASTM E1340-96, Standard Guide for Rapid Prototyping of Computerized Systems.[1]

IEEE Std 610.12-1990, IEEE Standard Glossary of Software Engineering Terminology.[2]

IEEE Std 730-1998, IEEE Standard for Software Quality Assurance Plans.

IEEE Std 730.1-1995, IEEE Guide for Software Quality Assurance Planning.

IEEE Std 828-1998, IEEE Standard for Software Configuration Management Plans.[3]

IEEE Std 982.1-1988, IEEE Standard Dictionary of Measures to Produce Reliable Software.

IEEE Std 982.2-1988, IEEE Guide for the Use of IEEE Standard Dictionary of Measures to Produce Reliable Software.

IEEE Std 1002-1987 (Reaff 1992), IEEE Standard Taxonomy for Software Engineering Standards.

IEEE Std 1012-1998, IEEE Standard for Software Verification and Validation.

IEEE Std 1012a-1998, IEEE Standard for Software Verification and Validation: Content Map to IEEE/EIA 12207.1-1997.[4]

IEEE Std 1016-1998, IEEE Recommended Practice for Software Design Descriptions.[5]

IEEE 830标准（英文版）(3)

3. Definitions

In general the definitions of terms used in this recommended practice conform to the definitions provided in IEEE Std 610.12-1990. The definitions below are key terms as they are used in this recommended practice.

3.1 contract: A legally binding document agreed upon by the customer and supplier. This includes the technical and organizational requirements, cost, and schedule for a product. A contract may also contain informal but useful information such as the commitments or expectations of the parties involved.

3.2 customer: The person, or persons, who pay for the product and usually (but not necessarily) decide the requirements. In the context of this recommended practice the customer and the supplier may be members of the same organization.

3.3 supplier: The person, or persons, who produce a product for a customer. In the context of this recommended practice, the customer and the supplier may be members of the same organization.

3.4 user: The person, or persons, who operate or interact directly with the product. The user(s) and the customer(s) are often not the same person(s).

IEEE 830标准（英文版）(4)

4. Considerations for producing a good SRS

This clause provides background information that should be considered when writing an SRS. This includes the following:

a) Nature of the SRS;
b) Environment of the SRS;
c) Characteristics of a good SRS;
d) Joint preparation of the SRS;
e) SRS evolution;
f) Prototyping;
g) Embedding design in the SRS;
h) Embedding project requirements in the SRS.

IEEE 830标准（英文版）(4.1)

4.1 Nature of the SRS

The SRS is a specification for a particular software product, program, or set of programs that performs certain functions in a specific environment. The SRS may be written by one or more representatives of the supplier, one or more representatives of the customer, or by both. Subclause 4.4 recommends both.

The basic issues that the SRS writer(s) shall address are the following:

a) *Functionality*. What is the software supposed to do?

b) *External interfaces*. How does the software interact with people, the system's hardware, other hardware, and other software?

c) *Performance*. What is the speed, availability, response time, recovery time of various software functions, etc.?

d) *Attributes*. What are the portability, correctness, maintainability, security, etc. considerations?

e) *Design constraints imposed on an implementation*. Are there any required standards in effect, implementation language, policies for database integrity, resource limits, operating environment(s) etc.?

The SRS writer(s) should avoid placing either design or project requirements in the SRS.

For recommended contents of an SRS see Clause 5.

IEEE 830标准（英文版）(4.2)

4.2 Environment of the SRS

It is important to consider the part that the SRS plays in the total project plan, which is defined in IEEE Std 610.12-1990. The software may contain essentially all the functionality of the project or it may be part of a larger system. In the latter case typically there will be an SRS that will state the interfaces between the system and its software portion, and will place external performance and functionality requirements upon the software portion. Of course the SRS should then agree with and expand upon these system requirements.

IEEE Std 1074-1997 describes the steps in the software life cycle and the applicable inputs for each step. Other standards, such as those listed in Clause 2, relate to other parts of the software life cycle and so may complement software requirements.

Since the SRS has a specific role to play in the software development process, the SRS writer(s) should be careful not to go beyond the bounds of that role. This means the SRS

a) Should correctly define all of the software requirements. A software requirement may exist because of the nature of the task to be solved or because of a special characteristic of the project.

b) Should not describe any design or implementation details. These should be described in the design stage of the project.

c) Should not impose additional constraints on the software. These are properly specified in other documents such as a software quality assurance plan.

Therefore, a properly written SRS limits the range of valid designs, but does not specify any particular design.

IEEE 830标准（英文版）(4.3)

4.3 Characteristics of a good SRS

An SRS should be

a) Correct;
b) Unambiguous;
c) Complete;
d) Consistent;
e) Ranked for importance and/or stability;
f) Verifiable;
g) Modifiable;
h) Traceable.

IEEE 830标准（英文版）(4.4)

4.4 Joint preparation of the SRS

The software development process should begin with supplier and customer agreement on what the completed software must do. This agreement, in the form of an SRS, should be jointly prepared. This is important because usually neither the customer nor the supplier is qualified to write a good SRS alone.

a) Customers usually do not understand the software design and development process well enough to write a usable SRS.

b) Suppliers usually do not understand the customer's problem and field of endeavor well enough to specify requirements for a satisfactory system.

Therefore, the customer and the supplier should work together to produce a well-written and completely understood SRS.

A special situation exists when a system and its software are both being defined concurrently. Then the functionality, interfaces, performance, and other attributes and constraints of the software are not predefined, but rather are jointly defined and subject to negotiation and change. This makes it more difficult, but no less important, to meet the characteristics stated in 4.3. In particular, an SRS that does not comply with the requirements of its parent system specification is incorrect.

IEEE 830标准（英文版）(4.5)

4.5 SRS evolution

The SRS may need to evolve as the development of the software product progresses. It may be impossible to specify some details at the time the project is initiated (e.g., it may be impossible to define all of the screen formats for an interactive program during the requirements phase). Additional changes may ensue as deficiencies, shortcomings, and inaccuracies are discovered in the SRS.

Two major considerations in this process are the following:

a) Requirements should be specified as completely and thoroughly as is known at the time, even if evolutionary revisions can be foreseen as inevitable. The fact that they are incomplete should be noted.

b) A formal change process should be initiated to identify, control, track, and report projected changes. Approved changes in requirements should be incorporated in the SRS in such a way as to

   1) Provide an accurate and complete audit trail of changes;
   2) Permit the review of current and superseded portions of the SRS.

IEEE 830标准（英文版）(4.6)

4.6 Prototyping

Prototyping is used frequently during the requirements portion of a project. Many tools exist that allow a prototype, exhibiting some characteristics of a system, to be created very quickly and easily. See also ASTM E1340-96.

Prototypes are useful for the following reasons:

a) The customer may be more likely to view the prototype and react to it than to read the SRS and react to it. Thus, the prototype provides quick feedback.

b) The prototype displays unanticipated aspects of the systems behavior. Thus, it produces not only answers but also new questions. This helps reach closure on the SRS.

c) An SRS based on a prototype tends to undergo less change during development, thus shortening development time.

A prototype should be used as a way to elicit software requirements. Some characteristics such as screen or report formats can be extracted directly from the prototype. Other requirements can be inferred by running experiments with the prototype.

IEEE 830标准（英文版）(5)

Table of Contents

1. Introduction
    1.1 Purpose
    1.2 Scope
    1.3 Definitions, acronyms, and abbreviations
    1.4 References
    1.5 Overview
2. Overall description
    2.1 Product perspective
    2.2 Product functions
    2.3 User characteristics
    2.4 Constraints
    2.5 Assumptions and dependencies
3. Specific requirements (See 5.3.1 through 5.3.8 for explanations of possible specific requirements. See also Annex A for several different ways of organizing this section of the SRS.)
Appendixes
Index

Figure 1—Prototype SRS outline

3. Specific requirements (1)

A.1 Template of SRS Section 3 organized by mode: Version 1

3. Specific requirements
- 3.1 External interface requirements
  - 3.1.1 User interfaces
  - 3.1.2 Hardware interfaces
  - 3.1.3 Software interfaces
  - 3.1.4 Communications interfaces
- 3.2 Functional requirements
  - 3.2.1 Mode 1
    - 3.2.1.1 Functional requirement 1.1
    - .
    - .
    - .
    - 3.2.1.*n* Functional requirement 1.*n*
  - 3.2.2 Mode 2
  - .
  - .
  - .
  - 3.2.*m* Mode *m*
    - 3.2.*m*.1 Functional requirement *m*.1
    - .
    - .
    - .
    - 3.2.*m*.*n* Functional requirement *m*.*n*
- 3.3 Performance requirements
- 3.4 Design constraints
- 3.5 Software system attributes
- 3.6 Other requirements

3. Specific requirements (2)

A.3 Template of SRS Section 3 organized by user class

3. Specific requirements
3.1 External interface requirements
3.1.1 User interfaces
3.1.2 Hardware interfaces
3.1.3 Software interfaces
3.1.4 Communications interfaces
3.2 Functional requirements
3.2.1 User class 1
3.2.1.1 Functional requirement 1.1
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.
3.2.1.*n* Functional requirement 1.*n*
3.2.2 User class 2
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.
.
3.2.*m* User class *m*
3.2.*m*.1 Functional requirement *m*.1
.
.
.
3.2.*m*.*n* Functional requirement *m*.*n*
3.3 Performance requirements
3.4 Design constraints
3.5 Software system attributes
3.6 Other requirements

3. Specific requirements (3)

A.4 Template of SRS Section 3 organized by object

3. Specific requirements
- 3.1 External interface requirements
  - 3.1.1 User interfaces
  - 3.1.2 Hardware interfaces
  - 3.1.3 Software interfaces
  - 3.1.4 Communications interfaces
- 3.2 Classes/Objects
  - 3.2.1 Class/Object 1
    - 3.2.1.1 Attributes (direct or inherited)
      - 3.2.1.1.1 Attribute 1
      - .
      - .
      - .
      - 3.2.1.1.*n* Attribute *n*
    - 3.2.1.2 Functions (services, methods, direct or inherited)
      - 3.2.1.2.1 Functional requirement 1.1
      - .
      - .
      - .
      - 3.2.1.2.*m* Functional requirement 1.*m*
    - 3.2.1.3 Messages (communications received or sent)
  - 3.2.2 Class/Object 2
  - .
  - .
  - .
  - 3.2.*p* Class/Object *p*
- 3.3 Performance requirements
- 3.4 Design constraints
- 3.5 Software system attributes
- 3.6 Other requirements

3. Specific requirements (4)

A.5 Template of SRS Section 3 organized by feature

3. Specific requirements
3.1 External interface requirements
3.1.1 User interfaces
3.1.2 Hardware interfaces
3.1.3 Software interfaces
3.1.4 Communications interfaces
3.2 System features
3.2.1 System Feature 1
3.2.1.1 Introduction/Purpose of feature
3.2.1.2 Stimulus/Response sequence
3.2.1.3 Associated functional requirements
3.2.1.3.1 Functional requirement 1
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.
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3.2.1.3.*n* Functional requirement *n*
3.2.2 System feature 2
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.
3.2.*m* System feature *m*
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.
.
3.3 Performance requirements
3.4 Design constraints
3.5 Software system attributes
3.6 Other requirements

本章小结

- 需求规格说明定义解决方案和需求，承载需求分析的成果
- 需求规格说明是一项复杂的活动，正确的文档写作要求准确的界定文档的特性
- 掌握文档模版的裁剪技巧和文档的写作技巧，可以帮助提高需求规格说明文档写作的能力

Q&A